

Name: _____ Date: _____

Period: _____

Exit Ticket — Answer independently

1. Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for all θ in the interval $[0, 2\pi]$.

Step 1 — Identify which quadrants have **negative** sine: **Quadrants III and IV (sine is negative below the x -axis).**

Step 2 — Reference angle: $\arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$

Step 3 — Write both solutions in $[0, 2\pi]$:

$$\theta = \frac{4\pi}{3} \text{ and } \theta = \frac{5\pi}{3}$$

Teacher Note

Q3 solution: $\pi + \frac{\pi}{3} = \frac{4\pi}{3}$. Q4 solution: $2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$. Both are in $[0, 2\pi]$.

2. Explain in one or two sentences why a trigonometric equation has *infinitely many* solutions when the domain is all real numbers.

Trigonometric functions are periodic, so they return to every output value once per period. Adding any integer multiple of the period 2π to a solution produces another valid solution, giving infinitely many solutions in an unrestricted domain.